9B Lesson 7

Bias, validity and organic farming

Year 9 Science · 9B Plant growth · Student Book pages 34 to 36

Name _____ Class _____ Date _____

Today I can:

- ☐ I can identify bias in graphs and arguments, explain where accidental bias comes from, and say whether a conclusion is valid.
- ☐ I can compare organic and intensive farming and explain biological control.

Key words 关键词

bias	偏差
balanced	客观全面的
systematic error	系统误差
random error	随机误差
sample	样本
valid	有效的
conclusion	结论
organic farming	有机农业
intensive farming	集约农业
biological control	生物防治
crop rotation	轮作

Command words 指令词

Draw 画出 · Explain 解释 · Give 给出 · Give a reason 给出理由 · Match 连线（配对） · State 说明（写出） · Suggest 提出（建议） · Tick 打勾 · Write 写出

● Core Everyone does this section.

1. Students added different numbers of drops of liquid fertiliser to bean seedlings and measured the mean increase in height after a week.

Number of drops of liquid fertiliser	2	4	6	8	10
Mean increase in height of seedlings (cm)	4.6	6.3	7.8	7.5	7.2

a) Tick one box to show a valid conclusion for this experiment.

- ☐ A Seedling height increases as the amount of fertiliser increases.
- ☐ B Too little or too much fertiliser can harm seedlings.
- ☐ C The more fertiliser that is added, the smaller the increase in growth.
- ☐ D Growth increases with more fertiliser up to a point, beyond which the amount of growth falls.

b) Give a reason for your choice.

c) State two variables that should be controlled to make this experiment valid.

2. Match each scientific term to the correct example. Draw a line.

systematic error	the stop clock was stopped late in one test during an experiment
random error	only data that support a particular view are quoted
valid experiment	a mass balance is not set to zero before the experiment
intentional bias	all other variables are controlled, so the dependent variable is only affected by the

	independent variable

3. Tick one box in each row to show whether the statement describes organic farming or intensive farming.

Statement	Organic	Intensive
mostly natural fertilisers such as manure		
many artificial pesticides		
large fields to make it easy to use big machines		
small fields with hedges and trees for predators of pests to live in		
crops are rotated so that clover adds nitrates to the soil		
the biggest yield from each field		

●● Stretch When Core is finished.

S1. A report in 2017 concluded that the mass of insects in Germany had decreased by 78% in 24 years. The data had been collected using traps that catch insects flying about a metre above the ground. Give two reasons why this conclusion may not be valid.

S2. Look at graph C on the slide (or page 34 of your book), which shows the monarch butterfly counts for 1997, 1999, 2001 and 2003 only.

a) Explain how this graph is biased.

b) Suggest who might create a graph like this, and explain your reasoning.

S3. Suggest which type of farm, organic or intensive, attracts more insect-eating birds. Explain your answer.

●●● Challenge Explain in full sentences.

C1. HT soybean is not killed by the herbicide glyphosate. Advantages: easier management of weeds; fewer sprayings needed than with other herbicides; glyphosate is not persistent. Disadvantages: expensive seed; with time, weeds unaffected by glyphosate become more common, so other herbicides may be needed; HT soybean may grow as a weed in other crops. Write a balanced paragraph to help a farmer decide whether to plant HT soybean.

C2. Two farms have the same area. One is organic and one is intensive.

a) Which farm produces the greater amount of crop? Explain your answer.

b) Explain why some people still choose to buy organic food, and state what the evidence says about whether organic food is healthier.

My thinking (any language 可用中文)

A website says: 'Scientists prove that organic food is healthier.' What would you want to know about the study before you believed this claim?

Word bank bias 偏差 · balanced 客观全面的 · systematic error 系统误差 · random error 随机误差 · sample 样本 · valid 有效的 · conclusion 结论 · organic farming 有机农业 · intensive farming 集约农业 · biological control 生物防治 · crop rotation 轮作

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Exit ticket 9B Lesson 7: Bias, validity and organic farming Name _____

1. What is bias?

2. Give one example of a systematic error.

3. Give one difference between organic farming and intensive farming.

Today I can: Goal 1 ☐ Goal 2 ☐